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Zoo501

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1. Name germ layers in animals 2

- Ectoderm
- Mesoderm
- Endoderm

2. Naïve 4

Uncommitted cell can be described as naive, meaning that it has received no instructions directing it along a particular developmental pathway.

3. Write two strategies of commitment 5

- **The inheritance of cytoplasmic determinants:** Cytoplasmic determinants are the molecules in cytoplasm that can help to determine cell fate.

For example: if a mother cell contains cytoplasmic determinants. Only one daughter inherited.

- **The perception of external inductive signals:** The process where one cell or group of cells changes developmental fate of another is termed induction. Two identical cells can follow alternative fates

4. Write modes of development commitment 6

- Autonomous specification
- Conditional specification
- Syncytial specification

5. Explain gametogenesis 11

The germ cells migrate to the gonads where they differentiate into gametes. The development of gametes is called gametogenesis. Spermatogenesis and oogenesis are both forms of gametogenesis, in which gamete cell produces haploid sperm and egg cells. The process of gametogenesis occurs in three steps. The 1st is Mitotic division in which germ cell grows. 2nd is two mitotic divisions to producing haploid daughter cells. 3rd is Differentiation of the haploid daughter cells to produce functional gametes.

6. Diff between germ and somatic cells with examples 11

Germ Cell: A reproductive cell of the body. Germ cells are egg in females and sperm in males.
Somatic Cell: Somatic cells are the cells in the body other than sperm and egg cells.

7. Note on frog 11

Frog is an amphibious organism it can live in both water and land. Frog mostly likes to live in water. During summer, when the water dries up, the frogs go inside the ground. They stay in the ground for months without eating and drinking and wait for the rain to come.
The life cycle of a frog consists of three stages: egg, larva, and adult. Frog grows, it moves through these stages in a process known as metamorphosis. In most species of frogs, fertilization is external. The male frog grabs the female's back and fertilizes the eggs as the female frog releases them. Rana pipiens lays 2500 eggs, and Rana catesbiana lays 20,000 eggs.

8. In which portion vegetal hemisphere present 12

The portion of egg that has large amount of yolk is known as vegetal hemisphere and it is present toward posterior side of the egg.

9. Explain diploblastic animal with example 18

Diploblastic animals are those who have ectoderm and endoderm, but no true mesoderm. These include the cnidarians (jellyfish and hydras)

10.Transcription in regulating 24

- **Differential gene transcription:** It regulates that which of the nuclear genes are transcribed into RNA.
- **Selective nuclear RNA processing:** It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.
- **Selective messenger RNA translation:** It regulates that which of the mRNAs in the cytoplasm become translated into proteins.
- **Differential protein modification:** It regulates that which proteins are allowed to remain or function in the cell.

11.Define fertilization generally 32-33

Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

12.What are microvilli? 40

Microfilaments are necessary for cell division and they are used to extend the egg surface into small projections called microvilli

13.Recognition of egg and sperm 41-42

- The chemo-attraction of sperm to egg by soluble molecules secreted by the egg
- The exocytosis of acrosomal release enzymes
- The binding of the sperm to the extracellular envelope of the egg
- The passing of the sperm through extracellular envelope
- Fusion of egg and sperm cell plasma membranes

14.Explain acrosomal reaction marine invertebrates 45

Acrosomal reaction has two components:

- Fusion of the acrosomal vesicle with the sperm plasma membrane
- Extension of the acrosomal process

In sea urchins, the acrosomal reaction is initiated by a fucose-containing polysaccharide in the egg jelly that binds to the sperm and allows calcium to enter in sperm head. The exocytosis of the acrosomal vesicle is caused by the calcium-mediated fusion of the acrosomal membrane with the adjacent sperm plasma membrane.

The second part of the acrosomal reaction involves the extension of the acrosomal process. This process arises through polymerization of globular actin molecules into actin filaments.

15.Fluidity of sperm 49-50

The fluidity of the sperm plasma membrane is altered by the removal of cholesterol by albumin proteins found in the female reproductive tract.

16.Mucopolysaccharides 59

Mucopolysaccharides released by the cortical granules produce an osmotic gradient that causes water to rush into the space between the plasma membrane and the vitelline envelope, causing the envelope to expand and become the fertilization envelope.

17.Early and late response from egg to sperm 62-63

Early responses:

Contact between sea urchin sperm and egg activates the two major blocks to polyspermy: the fast block, initiated by sodium influx into the cell, and the slow block, initiated by the intracellular release of calcium ions. The activation of all eggs appears to depend on an increase in the concentration of free calcium ions within the egg. Such an increase can occur in two ways: calcium ions can enter the egg from outside, or calcium ions can be released from the endoplasmic reticulum within the egg.

Late responses:

Shortly after the calcium ion levels rise in a sea urchin egg, its intracellular pH also increases. The rise in intracellular pH begins with a second influx of sodium ions, which causes a 1:1 exchange between sodium ions from the seawater and hydrogen ions from the egg. This loss of hydrogen ions causes the pH to rise. It is thought that the pH increase and the calcium ion elevation act together to stimulate new protein synthesis and DNA synthesis.

18.cytokinesis / karyokinesis 66-67

Karyokinesis means the process of division of the nucleus. The mechanical agent of division is mitotic spindle with microtubules composed of tubulin

Cytokinesis means the process of division of cytoplasm. The mechanical agent of cytokinesis is contractile ring of microfilaments made of actin

19.Which type of egg present in birds or chick 70

Telolecithal egg is present in birds. It is a type of egg that has large amount of yolk that is unevenly distributed in the cytoplasm

20.Composition of hyaline membrane 73-74

The hyaline layer is actually made up of two layers, an outer lamina made primarily of hyaline protein and an inner lamina composed of fibro-pellin proteins.

21.Type of gastrulation in snail 77-78

The snail stereoblastula is relatively small. Gastrulation in Snails is accomplished primarily by epiboly, wherein the micromeres at the animal cap multiply and "overgrow" the vegetal macromeres. Eventually, the micromeres will cover the entire embryo, leaving a small slit at the vegetal pole. In the typical cleavage of molluscs, cells divide to produce most of the larval structures, including a gland capable of producing a large shell.

22.Blastula formation in tunicates 79-80

Each successive division orients itself to this plane of symmetry, and the half-embryo formed on one side of the first cleavage plane is the mirror image of the half-embryo on the other side. The second cleavage is meridional, like the first, but unlike the first division, it does not pass through the centre of the egg. Rather, it creates two large anterior cells (the A and D blastomeres) and two smaller posterior cells (blastomeres B and C). Each side now has a large and a small blastomere. During the next three divisions, differences in cell size and shape highlight the bilateral symmetry of these embryos. At the 32-cell stage, a small blastocoel is formed, and gastrulation begins.

23.Gastrulation in tunicates 81-82

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm. About 4 5 hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding their apical margins and contracting near their vegetal margins. Pairs of blastomeres appear to lead this invagination into the centre of the embryo. This invagination forms a blastopore whose lips will become the mesodermal cells.

24.What are the three Axis of Body Foundation 83-84

Three axes are the foundations of the body: the anterior-posterior axis (head to tail or mouth to anus), the dorsal-ventral axis (back to belly), and the right-left axis (between the two lateral Sides of the body).

25.Cleavage type of sea urchin, snails, nematoda and tunicates 83-84

In all four invertebrates described here, cleavage is holoblastic. In the sea urchin, cleavage is radial; in the snail, spiral; in the tunicate, bilateral; and in the nematode, rotational.

26.Discoidal 85-86

The cytoplasm of the blastodisc becomes the embryo this type of meroblastic cleavage is called discoidal.

27.Meroblastic cleavage 85-86

The cell divisions do not completely divide the egg, so this type of cleavage is called meroblastic. The cytoplasm of the blastodisc becomes the embryo this type of meroblastic cleavage is called discoidal.

28.Syncytial layer of yolk 85-86

The first of these is the yolk syncytial layer (YSL). The YSL is formed at the ninth or tenth cell cycle, when the cells at the vegetal edge of the blastoderm fuse with the underlying yolk cell.

29.Embryonic shield 89

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the embryonic shield.

30.Sub Germinal cavity and its function 90-91

Between the blastoderm and the yolk is a space called the sub-germinal cavity. This space is created when the blastoderm cells absorb fluid from the albumin ("egg white") and secrete it between themselves and the yolk.

31.Secondary hypoblast form in avian 92-93

Sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

32.What is Henson node? 92-93

At the anterior end of the primitive streak is a regional thickening of cells called the primitive knot or Hensen's node.

33.Function of placenta 94-96

The placenta acts to provide oxygen and nutrients to the fetus, whilst removing carbon dioxide and other wasteproducts. It metabolises a number of substances and can release metabolic products into maternal and/or fetal circulations.

34.Umbilical cord 97-100

The narrow connecting stalk of extra embryonic mesoderm that links the embryo to the trophoblast eventually forms the vessels of the umbilical cord.

35.Note on different type of twins class in humans? 105-106

Human twins are classified into two major groups: monozygotic (one egg, or identical) twins and dizygotic (two egg, or fraternal) twins.

36.Amnion sac 105-106

This tissue forms the amnionic sac (or water sac), which surrounds the embryo with amnionic fluid and protects it from desiccation and abrupt movement.

37.Difference B/W identical and fraternal twins 105-106

Fraternal twins are the result of two separate fertilization events, whereas identical twins are formed from a single embryo whose cells somehow dissociated from one another.
Identical twins occur in roughly 0.25% of human births. About 33% of identical twins have two complete and separate chorions, indicating that separation occurred before the formation of the trophoblast tissue at day 5.

38.Fate of ectoderm 107-108

A portion of the dorsal ectoderm is specified to become neural ectoderm, and its cells become distinguishable by their columnar appearance. This region of the embryo is called the neural plate

39.Differentiation of the Neural Tube 107-108

In Primary Neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surfaceto form a hollow tube. In Secondary Neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube.

40.Neural grove formed in primary neurolation? 110

Shortly after the neural plate has formed, its edges thicken and move upward to form the neural folds, while a U-shaped neural groove appears in the center of the plate, dividing the future right and left sides of the embryo.

41.Define neural plate 110

The neural plate is a key developmental structure that serves as the basis for the nervous system. Opposite the primitive streak in the embryo, ectodermal tissue thickens and flattens to become the neural plate. The region anterior to the primitive knot can be generally referred to as the neural plate

42.MHP 111

In birds and mammals, the cells at the midline of the neural plate are called the medial hinge point (MHP) cells.

43.Spina bifida 112

Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida

44.Function of retinoic acid 115-118

In the retina, retinoic acid acts as a light-signaling neuromodulator and regulates gap junction-mediated coupling of retinal neurons. It has also been implicated in the control of sleep and circadian clock

45.Write the names of three vesicles formed in brain 115-118

- Prosencephalon (forebrain)
- Mesencephalon (midbrain)
- Rhombencephalon (hindbrain)

46.Difference b/w mentle and ventricular zone 119-120

As the cells adjacent to the lumen continue to divide, the migrating cells form a second layer around the original neural tube. This layer becomes progressively thicker as more cells are added to it from the germinal neuroepithelium. This new layer is called the mantle (or intermediate) zone, and the germinal epithelium is now called the ventricular zone (and, later, the ependyma).

47.Events in primary neurulation 119-120

- The internally positioned neural tube, which will form the brain and spinal cord

- The externally positioned epidermis of the skin.
- The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

48.What is neural crest write it's function? 125-126

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.The neural crest cells originate at the dorsal most region of the neural tube.The neural crest can be divided into four main functional (but overlapping) domains.
The cranial (cephalic) neural crest, whose cells migrate dorsolaterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia,and connective tissues of the face.

49.Factors in myogenesis 131-132

Paracrine factors instruct the Myotome cells to become muscles by inducing them to synthesize the MyoD protein. Factors from the surrounding environment induce the Pax3 transcription factor. In the absence of other inhibitory transcription factors (such as those found in the sclerotome cells), Pax3 then activates the genes encoding two muscle-specific transcription factors, Myf5 and MyoD. MyoD and Myf5 belong to a family of transcription factors called the myogenic bHLH (basic helix-loop-helix) proteins (sometimes also referred to as the MyoD family).

50.How muscle cell fusions occur and steps? 133-134

Muscle cell fusion begins when the myoblasts leave the cell cycle. As long as particular growth factors (particularly fibroblast growth factors) are present, the myoblasts will proliferate without differentiating.

Steps:

When these factors are depleted, the myoblasts stop dividing, secrete fibronectin onto their extracellular matrix, and bind to it through 5β1 integrin, their major fibronectin receptor. The second step is the alignment of the myoblasts together into chains.
The third step is the cell fusion event itself. As in most membrane fusions, calcium ions are critical, and fusion can be activated by calcium ionophores, such as A23187, that carry calcium ions across cell membranes. Fusion appears to be mediated by a set of metalloproteinases called meltrins. These proteins were discovered during a search for myoblast proteins that would be homologous to fertilin, a protein implicated in sperm-egg membrane fusion.

51.How myotubes forms 133-134

The myotome cells producing the myogenic bHLH proteins are the myoblasts- committed muscle cell precursors. These cells align together and fuse to form the multinucleated myotubes, a characteristic of muscle tissue

52.Endochondral ossification 135-136

The mesenchymal cells differentiate into cartilage, and this cartilage is later replaced by bone. The process by which a cartilage intermediate is formed and replaced by bone cells is called endochondral ossification

53.Three distinct lineage of skeleton 135-136

- The somites generate the axial skeleton.
- The lateral plate mesoderm generates the limb skeleton.

- The cranial neural crest gives rise to the branchial arch and craniofacial bones and cartilage.

54.Camptomelic dysplasia 139-142

Mutations of the SOX9 gene cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body.

55.Which factor is responsible for cartilage specifications? 139-142

Pax1 and Scleraxis are the transcription factors are thought to activate cartilage-specific genes.

56.What are osteoclasts? 143-144

Osteoclasts are probably derived from the same precursors as macrophage blood cells, and they dissolve both the inorganic and the protein portions of the bone matrix. Each osteoclast extends numerous cellular processes into the matrix and pumps out hydrogen ions onto the surrounding material, thereby acidifying and solubilizing it. The blood vessels also import the blood-forming cells that will reside in the marrow for the duration of the organism's life. The number and activity of osteoclasts must be tightly regulated. If there are too many active osteoclasts, too much bone will be dissolved, and osteoporosis will result. Conversely, if not enough osteoblasts are produced, the bones are not hollowed out for the marrow, and osteopetrosis results.

57.Blastula in snail?

58.Define Senscor

59.Movement name in gastrulation in invertebrate

60.

warriors